
Hidden-State Non-Recoverability and Action-Relevance Certificates for Agent Systems

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Abstract

Auditing a deployed language-model agent requires two lower certificates: one that the visible transcript cannot recover the operative hidden state, and one that the hidden state still affects the next action. We introduce a dual-certificate protocol for $\varepsilon_{\text{unrec}}^{\text{LB}}(t) \leq H(S_t | \tilde{T}_t)$ and $\delta_{\text{act}}^{\text{LB}}(t) \leq I(S_t; A_t | \tilde{T}_t)$. The first certificate is obtained from finite hidden-state witnesses or closed-form finite-dimensional witness models; the second is obtained from admissible replay, intervention, or proxy variables by conditional data processing. A structural cut-set budget remains useful, but only as an upper envelope on how much hidden state can escape the trace. The key audit failure is therefore a gap: citation-level or transcript-level audit can pass while a decision-relevant state variable remains unrecoverable. We instantiate the framework on multi-agent private-report edges, diffusion-LM denoising latents, and ReAct scratchpads. The strongest realized-action result is multi-agent: swapping a worker’s private report in a controller changes the API-action distribution by 0.901 bits (95% CI [0.873, 0.928]). LLaDA perturbations rise at the final binding step (0.110 bits, 95% CI [0.052, 0.234]). ReAct provides a conservative soft-policy diagnostic: under the same topology, controlled replay separates dormant calculator tasks from active planning tasks in tool-token probabilities (0.0163 bits, 95% CI [0.0124, 0.0208]), without changing argmax tool counts. These studies expose states that are unrecoverable from the transcript yet still action-relevant; independent state-witness non-recoverability is demonstrated in closed-form laboratories and synthetic ground-truth calibration.

1 Introduction

Consider a controller agent that receives a worker’s private report and then chooses between a calculator and a search API. The report can pass a transcript-level audit—it need not leak the API label verbatim—while a counterfactual swap of the worker report changes the controller’s realized action distribution by 0.901 bits. This is the audit failure mode: the visible transcript can look adequate while an unobserved routing or evidence state remains both unrecoverable and decision-relevant. A weaker version appears in ReAct scratchpads: controlled replay changes the soft tool-token distribution without changing argmax tool counts. That is useful as a policy-readout diagnostic, but it is not the headline behavioral result. Identifying which hidden channels are merely present, and which ones actually drive decisions, is the audit task.

How much internal state is unrecoverable from the trace, and how much of that state drives the next action, are distinct quantities. Treating them as a single “hiddenness” judgment produces something neither computable nor diagnostic: a system with unrecoverable but dormant hidden state is different from one whose unrecoverable state changes the action distribution, even when the visible transcript looks equally complete.

We propose a dual-certificate protocol that answers each question separately. The non-recoverability certificate $\varepsilon_{\text{unrec}}^{\text{LB}}(t)$ lower-bounds residual hidden-state uncertainty $H(S_t | \tilde{T}_t)$ by measuring finite witnesses of the hidden state or by evaluating closed-form finite-dimensional witness models. The action-relevance certificate $\delta_{\text{act}}^{\text{LB}}(t)$ lower-bounds $I(S_t; A_t | \tilde{T}_t)$ from admissible replay, intervention, or proxy measurements using conditional data processing. A structural cut-set budget $\varepsilon_{\text{state}}^{\text{UB}}$ is still reported when architecture access is available, but its role is an upper envelope on residual capacity, not the main NeurIPS contribution. The central audit question is whether *both*

$$H(S_t | \tilde{T}_t) \geq \varepsilon_{\text{unrec}}^{\text{LB}}(t), \quad I(S_t; A_t | \tilde{T}_t) \geq \delta_{\text{act}}^{\text{LB}}(t)$$

hold with nontrivial lower bounds.

The dynamic axis can also be indexed. Instead of asking only whether residual state matters, an auditor can ask *where* it matters: which module, denoising step, or communication edge carries decision relevance. This turns the certificate from a scalar diagnostic into an activation profile over the hidden-state geometry of the deployment.

Conditioning on $\tilde{T}_t$ is part of the protection, not a nuisance term. If a model writes a steganographic or compressed rationale into the recorded trace, that information is in $\tilde{T}_t$ and is conditioned away by $I(S_t; A_t | \tilde{T}_t)$. The dynamic certificate measures residual decision relevance beyond the logged trace; it does not count hiddenness that has already been made visible to the audit record.

Contributions. First, we separate transcript recoverability from residual action relevance as two distinct audit questions. Second, we give dual lower certificates ($\varepsilon_{\text{unrec}}^{\text{LB}}, \delta_{\text{act}}^{\text{LB}}$) for those questions. Third, under explicit audit-abstraction assumptions, we prove certificate validity: finite hidden-state witnesses lower-bound $H(S_t | \tilde{T}_t)$, and admissible probes lower-bound $I(S_t; A_t | \tilde{T}_t)$. Fourth, we give closed-form or computable audit instances for diffusion trajectories, retrieval-style hidden source selection, and nested agent architectures. An auxiliary cut-set budget is used only for logging ablations and deployment reporting.

We instantiate the protocol on three audit geometries: a ReAct scratchpad module, a diffusion-LM denoising trajectory, and a multi-agent private-report edge. These real-system diagnostics instantiate the structural budget and action-relevance certificate. Appendix calibrations cover a synthetic ground-truth setting where an independent hidden-state witness makes $\varepsilon_{\text{unrec}}^{\text{LB}}$ explicit and the structural budget matches true hidden-state entropy.

2 Related Work

Probing, patching, and causal abstraction. Causal abstraction [Geiger et al., 2021], latent-knowledge elicitation [Burns et al., 2022], nullspace projection [Ravfogel et al., 2020], and amnesic probing [Elazar et al., 2021] are natural sources of proxy-style evidence: when they expose a probe variable $Z_t = \varphi(S_t)$ and the required conditional assumptions hold, its MI with action can be interpreted through the proxy certificate. Causal tracing/ROME [Meng et al., 2022], representation engineering [Zou et al., 2023], and activation addition [Turner et al., 2023] provide intervention-style evidence when the patch affects action only through S_t .

Black-box audit and network information theory. Property testing [Goldreich et al., 1998] and black-box safety auditing [Casper et al., 2024, Su et al., 2024] study what can be inferred from outputs alone. The distinction here is that output-only access can support behavioral tests, but it does not by itself provide a lower bound on δ_{act} ; $\varepsilon_{\text{state}}^{\text{UB}}$ remains computable from topology when structural access is available. The structural budget proof applies a cut-set upper bound [El Gamal and Kim, 2011] to the time-unrolled DAG (full derivation in Appendix A).

Diffusion language-model agents. LLaDA is a large-scale masked diffusion language model with bidirectional denoising and instruction-following [Nie et al., 2025]. Recent agent work further studies diffusion LMs in multi-step decision and tool-use workflows, including matched comparisons to autoregressive agents [Zhen et al., 2026]. Intermediate denoising latents in these systems are a natural hidden channel for the dynamic certificate.

84 3 Setup and Audit Regime

85 Standard information-theoretic notation follows Cover and Thomas [2006]. All logarithms are base
86 2; entropy and mutual information are reported in bits.

87 At step t : *full trace* T_t (all intermediate activations), *recorded trace* $\tilde{T}_t \subseteq T_t$ (auditor-visible),
88 *operative state* S_t (internal computation), *action* A_t (next token), *unrecorded sources* U_t (exogenous
89 inputs not in $\tilde{T}_t$).

90 3.1 Two Core Audit Quantities

91 **Definition 1** (Core Audit Quantities). *The two quantities targeted by the dual certificate are:*

- 92 • **Hidden-state non-recoverability:** $\varepsilon_{\text{unrec}} := H(S_t \mid \tilde{T}_t)$, the residual entropy of the operative
93 state given the visible trace.
- 94 • **Residual decision relevance:** $\delta_{\text{act}} := I(S_t; A_t \mid \tilde{T}_t)$, the mutual information between
95 operative state and next action given the visible trace.

96 Neither quantity is directly computable in deployment: $\varepsilon_{\text{unrec}}$ requires recovering the full internal-state
97 distribution, and δ_{act} requires observing S_t . The dual-certificate framework targets:

- 98 • a non-recoverability lower bound $\varepsilon_{\text{unrec}}^{\text{LB}} \leq \varepsilon_{\text{unrec}}$, computed from hidden-state witnesses or
99 closed-form finite-dimensional models;
- 100 • an action-relevance lower bound $\delta_{\text{act}}^{\text{LB}} \leq \delta_{\text{act}}$, estimated via admissible replay, intervention,
101 or proxy probes (§5).

102 The reported audit pair is $(\varepsilon_{\text{unrec}}^{\text{LB}}, \delta_{\text{act}}^{\text{LB}})$. When architecture access is available, we also report a
103 structural budget $\varepsilon_{\text{state}}^{\text{UB}} \geq H(S_t \mid \tilde{T}_t)$ (§4).

104 Neither quantity is identifiable from behavioral observations alone: different causal graphs can
105 produce identical joint distributions over $(\tilde{T}_t, A_t)$ pairs, so observing outputs does not fix $H(S_t \mid \tilde{T}_t)$
106 or $I(S_t; A_t \mid \tilde{T}_t)$. The non-recoverability certificate addresses the first by measuring or analytically
107 computing hidden-state witnesses; the action-relevance certificate addresses the second by measuring
108 variables whose influence on action must pass through the operative state. The structural budget
109 bounds how large the first quantity could be under the deployment topology.

110 3.2 Audit Abstraction Assumptions

111 The certificates are claims about an audit abstraction, not about arbitrary deployment internals. The
112 abstraction declares the variables the auditor will treat as state, transcript, action, witnesses, probes,
113 and optional cut channels.

114 **Assumption 1** (Deployment abstraction). *For the audited step t , the deployment induces a joint*
115 *distribution over $(S_t, \tilde{T}_t, A_t)$ at a declared finite audit resolution. Any continuous internal variable*
116 *used in a certificate is first mapped to a finite-noise, finite-dimensional witness or probe.*

117 **Assumption 2** (Witness and probe admissibility). *Each non-recoverability witness $W_t^{(j)}$ is generated*
118 *from S_t by a known channel. Each action-relevance probe $X_t^{(j)}$ affects the next action only through*
119 *the operative state after conditioning on the transcript: $X_t^{(j)} \rightarrow S_t \rightarrow A_t$ given $\tilde{T}_t$.*

120 **Assumption 3** (Admissible cut budget). *When a structural upper budget is reported, the deployment*
121 *abstraction also provides a time-unrolled graph whose unlogged cuts separate unrecorded sources*
122 *from S_t , together with per-edge capacity budgets that upper-bound the conditional information*
123 *crossing each unlogged channel.*

124 **Theorem 1** (Dual certificate validity). *Under the deployment and probe assumptions, let $\varepsilon_{\text{unrec}}^{\text{LB}}(t)$*
125 *be any lower confidence bound on $\max_j I(W_t^{(j)}; S_t \mid \tilde{T}_t)$, and let $\delta_{\text{act}}^{\text{LB}}(t)$ be any lower confidence*
126 *bound on $\max_j I(X_t^{(j)}; A_t \mid \tilde{T}_t)$. Then*

$$H(S_t \mid \tilde{T}_t) \geq \varepsilon_{\text{unrec}}^{\text{LB}}(t), \quad I(S_t; A_t \mid \tilde{T}_t) \geq \delta_{\text{act}}^{\text{LB}}(t).$$

127 Thus the certified state is unrecoverable from the transcript, yet still action-relevant whenever both
128 lower bounds are nonzero.

129 *Proof.* The first inequality follows from conditional data processing and $I(W_t^{(j)}; S_t | \tilde{T}_t) \leq H(S_t |$
130 $\tilde{T}_t)$. The second follows from conditional data processing under $X_t^{(j)} \rightarrow S_t \rightarrow A_t$ given $\tilde{T}_t$.
131 Replacing population quantities by valid lower confidence bounds preserves both inequalities. $\square$

132 **Remark 1** (Complementarity and Inheritance). *By the data processing inequality, $\delta_{act} \leq \varepsilon_{state}$. The*
133 *certificates inherit this ordering at their respective bound directions but remain independent axes of*
134 *audit evidence:*

- 135 • If $\varepsilon_{state}^{UB} = 0$ (fully logged architecture), then $\varepsilon_{state} = 0$ and $\delta_{act} = 0$; any valid lower bound
136 satisfies $\delta_{act}^{LB} = 0$.
- 137 • If $\delta_{act}^{LB} = 0$ under the best available probe, hidden state may still exist: ε_{state}^{UB} can be large
138 while the current task does not activate the hidden capacity (dormant unrecoverable state).
139 The contrapositive ($\delta_{act}^{LB} > 0 \Rightarrow \varepsilon_{state}^{UB} > 0$) follows from the DPI ordering.

140 The audit interpretation is therefore two-dimensional: deployment teams must report both the
141 residual channel budget and the strongest admissible behavioral probe, not collapse them into a
142 single hiddenness score.

143 3.3 Audit Access Modes

- 144 1. *Structural access.* The auditor has the deployment architecture, logging manifest, and
145 protocol budgets. This is sufficient to compute ε_{state}^{UB} from topology (§4) but gives no
146 dynamic lower bound.
- 147 2. *Controlled replay access.* The auditor can rerun the same system under wild and replayed
148 states while holding the visible trace fixed. This supports replay certificates when the replay
149 mechanism affects the action only through missing state recovery.
- 150 3. *Proxy or intervention access.* The auditor can read a proxy $Z_t = f(S_t)$ or perturb a hidden
151 module. These support conditional-DPI lower bounds on δ_{act} (§5).

152 All three modes are instantiated in the controlled ReAct experiments below. Paraphrase- or format-
153 based black-box replay is not used as evidence for internal decision relevance.

154 4 Structural Budget: Untraced-Channel Cut Capacity

155 The structural budget upper-bounds $\varepsilon_{unrec} = H(S_t | \tilde{T}_t)$ from deployment topology. Information not
156 in the visible trace that can influence S_t must travel through unlogged channels. Bounding those
157 channels' worst-case capacity bounds the hidden-state entropy.

158 4.1 Time-Unrolled Deployment Graph and Component Budgets

159 Let $G_t = (V_t, E_t)$ be the time-unrolled DAG of the deployment up to step t . Nodes are state updates,
160 tool calls, memory reads/writes, messages; edges are information channels. Three object classes:
161 unrecorded sources U_t (retrieval results, external memory, unlogged messages), the visible trace $\tilde{T}_t$,
162 and the operative state S_t . Let $\mathcal{C}_{unlogged}$ be all reachable edge cuts separating U_t from S_t .

163 A set of edges is *software-orthogonal* if, conditioned on $\tilde{T}_t$, the channel outputs factor: $p(\{y_e\}_{e \in E'} |$
164 $\{x_e\}, \tilde{T}_t) = \prod_{e \in E'} p(y_e | x_e, \tilde{T}_t)$. This holds when unlogged channels use independent API calls or
165 separate memory regions (see Appendix A for non-orthogonal remediation).

166 **Lemma 1** (Additive decomposition). *Under software orthogonality with per-edge budgets c_e , the*
167 *induced cut capacity decomposes: $C_{cut}(\Omega) \leq \sum_{e \in E(\Omega, \Omega^c)} c_e$.*

168 When software orthogonality fails (for example, when two unlogged channels share a hidden state
169 variable) the per-edge sum $\sum_{e \in E(\Omega, \Omega^c)} c_e$ still upper-bounds $C_{cut}(\Omega)$ by sub-additivity of mutual
170 information across coupled channels: $I(X_\Omega; Y_{\Omega^c} | \tilde{T}_t, X_{\Omega^c}) \leq \sum_e I(X_e; Y_e | \tilde{T}_t) \leq \sum_e c_e$.

Orthogonality is required for *tightness* of the additive decomposition, not for validity of the upper bound (formal treatment in Appendix A).

Per-edge budgets c_e are read from interface specifications: $K \log |\mathcal{V}|$ for a K -token text channel over vocabulary $\mathcal{V}$, $d \log Q$ for a d -dimensional quantized state, $\log |\Omega_{\text{states}}|$ for a discrete scheduler. The discretization parameter Q is an auditor-side choice (we use $Q = 256$, 8-bit audit discretization throughout); tighter or coarser discretizations shift the bound proportionally without changing the structural claim. All per-edge bounds are worst-case (maximum-entropy) bounds; tighter empirical bounds only improve the certificate.

Proposition 1 below has three terms. $\varepsilon_{\text{nominal}} = H(S_t | T_t)$ is the residual uncertainty after a complete internal trace; for a fully instrumented software agent this is zero. The cut term budgets information that can still enter S_t through unlogged interfaces. The directed-MI notation prices that worst-case flow formally; the implemented certificate needs only per-edge budgets and a min-cut on the unlogged subgraph.

Proposition 1 (Static Certificate via Cut-Set Bound). *For any cut Ω separating U_t from S_t with induced capacity $C_{\text{cut}}(\Omega) = \sup_{p(X_\Omega | \tilde{T}_t)} I(X_\Omega \rightarrow Y_{\Omega^c} | \tilde{T}_t, X_{\Omega^c})$,*

$$H(S_t | \tilde{T}_t) \leq \underbrace{H(S_t | T_t)}_{\varepsilon_{\text{nominal}}} + \min_{\Omega} C_{\text{cut}}(\Omega).$$

Setting $\varepsilon_{\text{state}}^{\text{UB}} := \varepsilon_{\text{nominal}} + \min_{\Omega} C_{\text{cut}}(\Omega)$ gives $\varepsilon_{\text{state}} \leq \varepsilon_{\text{state}}^{\text{UB}}$.

Here X_Ω denotes the input signals entering edges crossing the cut, Y_{Ω^c} the corresponding output signals, and X_{Ω^c} the boundary state on the sink side (formal definitions in Appendix A).

Corollary 1 (Additive form). *Under software orthogonality, $\varepsilon_{\text{state}}^{\text{UB}} = \varepsilon_{\text{nominal}} + \min_{C \in \mathcal{C}_{\text{unlogged}}} \sum_{e \in C} c_e$, a discrete min-cut on the unlogged subgraph.*

Proof sketch: Chain-rule gives $H(S_t | \tilde{T}_t) = H(S_t | T_t) + I(S_t; M_t | \tilde{T}_t)$ where $M_t = T_t \setminus \tilde{T}_t$. Every path from U_t to S_t crosses a cut Ω , so M_t is d -separated from S_t by the cross-cut signals; DPI yields $I(S_t; M_t | \tilde{T}_t) \leq C_{\text{cut}}(\Omega)$. Minimizing over cuts and substituting Lemma 1 completes the proof (full derivation in Appendix A).

Corollary 2 (Autoregressive zero-cut). *If the system is a strict autoregressive core and $\tilde{T}_t$ includes the full context window, then $\mathcal{C}_{\text{unlogged}} = \emptyset$, every cut has zero induced capacity, and $\varepsilon_{\text{state}}^{\text{UB}} = 0$.*

5 Dynamic Certificate: Decision Relevance via Conditional DPI

The dynamic certificate targets $\delta_{\text{act}} := I(S_t; A_t | \tilde{T}_t)$. Since S_t is unobservable, we specify probes whose measured variables satisfy $X_t \rightarrow S_t \rightarrow A_t$ given $\tilde{T}_t$. Conditional DPI supplies the correctness argument: $I(X_t; A_t | \tilde{T}_t) \leq \delta_{\text{act}}$.

Proposition 2 (Probe Certificates from Conditional DPI). *For any admissible probe variable X_t with $X_t \rightarrow S_t \rightarrow A_t$ given $\tilde{T}_t$, the probe measurement $I(X_t; A_t | \tilde{T}_t)$ is a valid lower-bound certificate for δ_{act} .*

Replay certificates. Let $R_t \in \{\text{wild}, \text{replay}\}$ indicate whether a missing state fragment is reconstructed under the same visible trace $\tilde{T}_t$ (controlled re-execution, not paraphrase variation). Then $I(R_t; A_t | \tilde{T}_t) = \text{JS}(P_{\text{wild}}, P_{\text{replay}} | \tilde{T}_t) \leq \delta_{\text{act}}$. Discrete action spaces admit direct empirical JS estimation over either realized actions or model-reported action probabilities; the experiments below state which is used.

Intervention certificates. Let ξ_{hidden} be an exogenous perturbation to a suspected unlogged module (cache, memory buffer, scratchpad). If $\xi_{\text{hidden}} \rightarrow S_t \rightarrow A_t$ given $\tilde{T}_t$, then $I(\xi_{\text{hidden}}; A_t | \tilde{T}_t) \leq \delta_{\text{act}}$. Discrete action spaces admit direct conditional action-distribution shift estimates, such as JS divergence; for continuous action spaces, InfoNCE [van den Oord et al., 2018] or MINE [Belghazi et al., 2018] can be used. Requires gray-box perturbation access; the most direct causal probe of the three.

Logging level	$\varepsilon_{\text{state}}^{\text{UB}}$ (bits)	Bottleneck
ReAct, output only	16,464	scratchpad_read (16384) + router_logits (80)
ReAct, +log router	16,384	scratchpad_read (16384)
ReAct, +log scratchpad	0	none
ReAct, full instrumentation	0	none
LLaDA-8B-Instruct ($K=128$)	32,768	denoising interface
Multi-agent, all logged	0	none
Multi-agent, peer report unlogged	32,768	peer_report $\rightarrow S_t$

Table 1: Structural budget geometry. Adding ReAct logs monotonically removes residual cut capacity; the same min-cut analysis localizes capacity in a scratchpad module, denoising interface, or peer-report edge. The LLaDA bound is independent of K because repeated reuse of the same bounded latent interface does not increase the one-step information bottleneck.

Proxy certificates. Let $Z_t = f(S_t)$ be a readable coarsening of the operative state (tool-logit projection, attention summary). Since $Z_t \leftarrow S_t \rightarrow A_t$ given $\tilde{T}_t$, conditional DPI yields $I(Z_t; A_t | \tilde{T}_t) \leq \delta_{\text{act}}$. We estimate this quantity by fitting predictors $p(A_t | \tilde{T}_t)$ and $p(A_t | \tilde{T}_t, Z_t)$ and taking their cross-entropy gap, $\hat{I}(Z_t; A_t | \tilde{T}_t)$.

Aggregation. A single certificate class can miss some causal paths. Define $\delta_{\text{act}}^{\text{LB}} := \max\{I(R_t; A_t | \tilde{T}_t), I(\xi_{\text{hidden}}; A_t | \tilde{T}_t), I(Z_t; A_t | \tilde{T}_t)\}$. By monotonicity of the maximum, the aggregate remains a valid lower bound on δ_{act} .

Activation profiles. The scalar certificate is the maximum over an indexed family of admissible probes. Keeping the index gives an *activation profile*

$$\Delta(j) := I(X_t^{(j)}; A_t | \tilde{T}_t),$$

where j may denote a scratchpad field, a denoising step, an activation layer, or a communication edge. The structural budget identifies where residual capacity can reside; the activation profile identifies where that capacity is behaviorally used. The experiments below keep this algebra fixed while changing the index: module in ReAct, time in the diffusion LM, and communication edge in the multi-agent setting.

6 Empirical Diagnostics

The static ablation (§6.1) computes the residual bit budget from topology. ReAct intervention and controlled replay (§6.2) show the same unlogged scratchpad dormant on calculator tasks and active on planning tasks. The LLaDA and multi-agent probes index $\Delta(j)$ by denoising step and peer-report edge, respectively (§6.3–6.4). Read-only proxy and synthetic ground-truth calibrations are in Appendices C–D. All pipelines run on a single M4 Max workstation.

6.1 Structural Budget: Min-Cut and Logging Ablation

We extract the time-unrolled DAG from Qwen2.5-7B-Instruct ReAct execution traces on tool-selection tasks (Fig. 1, left). Per-edge budgets c_e are assigned from interface specifications (§4); the min-cut on unlogged edges gives $\varepsilon_{\text{state}}^{\text{UB}}$. Varying logging levels recomputes the min-cut; Table 1 reports the stepwise reduction from 16,464 to 0 bits. The dominant bottleneck is the scratchpad-read path (16,384 bits). The same static computation also supplies the coordinate system for the diffusion-LM and multi-agent dynamic checks that follow.

These cuts put the later dynamic probes on a common map: a scratchpad module in ReAct, a denoising interface in LLaDA, and a peer-state edge in the multi-agent system.

6.2 Dynamic Causal Certificates: Intervention and Controlled Replay

We separate the two axes with a dormant/active task split under one topology. The unlogged scratchpad is present in both cases: irrelevant for calculator tasks, necessary for planning tasks.

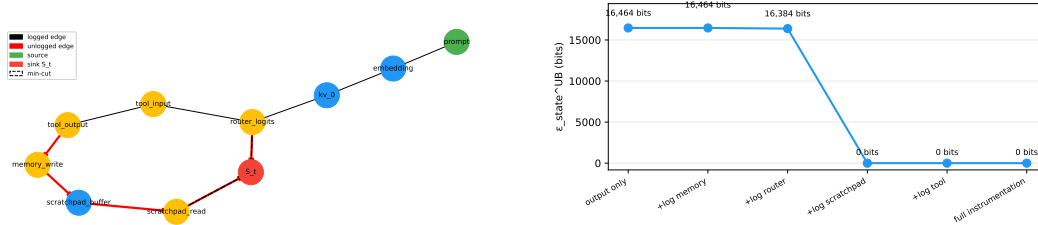


Figure 1: Left: Time-unrolled DAG extracted from Qwen2.5-7B-Instruct ReAct execution (logical edges omitted for clarity). Red edges are unlogged. Dashed edges form the min-cut. Right: Logging ablation curve. Adding instrumentation at the bottleneck drops $\varepsilon_{\text{state}}^{\text{UB}}$ stepwise to zero.

Task	Perturbation	JS (bits; nats)	95% CI (bits)
Calculator (dormant)	mask 0.7	0.0000; 0.0000	[0, 0]
Planning (active)	mask 0.7	0.0215; 0.0149	[0.0094, 0.0410]
Calculator (dormant)	replace 1.0	0.0000; 0.0000	[0, 0]
Planning (active)	replace 1.0	0.0055; 0.0038	[0.0016, 0.0131]

Table 2: Dynamic intervention certificate under the same unlogged-scratchpad topology. Dormant calculator tasks show no detectable scratchpad influence; active planning tasks show positive action-distribution shift.

Intervention perturbs the scratchpad (masking or replacement); controlled replay removes it entirely while holding $\tilde{T}_t$ fixed. Table 2 reports JS divergence over realized tool actions. Table 3 reports JS divergence over the model’s soft tool-token probability distribution; for the headline replay contrast, the argmax tool counts are unchanged, so the evidence is a policy-probability shift rather than a realized-tool flip. Both certificates are zero on dormant tasks and positive on active tasks. The topology, and therefore $\varepsilon_{\text{state}}^{\text{UB}}$, is unchanged.

The pair of tables isolates the transition from high- ε /low- δ to high- ε /high- δ ; the hidden channel is structurally present in both task splits, but it becomes action-relevant only on planning tasks.

Under this fixed topology, the unlogged scratchpad contributes 16,384 bits to $\varepsilon_{\text{state}}^{\text{UB}}$ in both task splits, matching the min-cut from the extracted ReAct DAG. Intervention and replay are zero on calculator tasks and positive on planning tasks. Structural unrecoverability is set by the deployment boundary and logging policy; residual decision relevance depends on which hidden channels the task uses.

6.3 Temporal Certificate Analysis on a Diffusion LM

ReAct gives a module-level contrast: the scratchpad is either used or not used. A diffusion LM gives a temporal contrast. The hidden channel exists throughout the K -step denoising trajectory, but its decision relevance need not be constant across steps.

We use LLaDA-8B-Instruct with $K=10$ denoising steps on tool-selection prompts. A Gaussian perturbation ($\sigma=5.0$) is applied to layer-1 activations at steps $\{2, 4, 6, 8, 10\}$; a late-layer perturbation is tracked as a broad sensitivity control. Each point uses $n=20$ trajectories and a trajectory-block bootstrap with 1,000 resamples.

The pattern is late-binding. Steps 2–8 remain near the floor for the target layer (0.001–0.006 bits), while step 10 rises to 0.110 bits with 95% CI [0.052, 0.234]. The control layer also rises at step 10 (0.092 bits, CI [0.043, 0.186]), so the sensitivity is global rather than confined to the chosen layer. LLaDA carries hidden capacity throughout denoising; it becomes action-relevant when the denoising state binds to a discrete tool choice.

6.4 Multi-Agent Private-Report Intervention

The multi-agent setting changes the index again. Here the controller delegates evidence gathering to a worker, so the worker’s private report is not an auxiliary scratchpad but an epistemic edge in

Task	Replay condition	JS (bits; nats)	95% CI (bits)
Calculator (dormant)	scratchpad removed	0.0000; 0.0000	[0, 0]
Planning (active)	scratchpad removed	0.0163; 0.0113	[0.0124, 0.0208]

Table 3: Controlled replay certificate over soft tool-token probability distributions. Wild episodes use the full hidden scratchpad; replay episodes use the same visible trace with the scratchpad removed. The dormant-to-active contrast mirrors the intervention result, although the headline replay contrast does not change argmax tool counts.

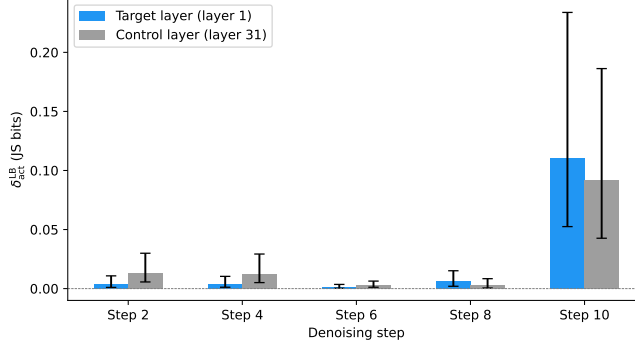


Figure 2: Temporal activation profile for LLaDA-8B-Instruct. Early denoising steps have low dynamic certificates; the final action-forming step has a much larger certificate for both target and control layers, indicating late-binding decision relevance rather than a module-local scratchpad effect.

the decision graph. The worker sees a benchmark item and writes a short report; the controller sees only that report and chooses one API from {calculator, search}. The task pool combines 200 GSM8K arithmetic items [Cobbe et al., 2021] and 200 HotpotQA multi-hop fact questions [Yang et al., 2018] with seed 42.

The main intervention replaces the worker report with a report from the opposite task class while holding the controller protocol fixed. A neutral replay removes the report content, and a same-class shuffle replaces it with another report from the same task class. An oracle-tag condition directly names and flips the API label to measure the binary ceiling. For each task and condition, the controller is sampled $k=8$ times; labels are parsed as {calculator, search, invalid}; and $\text{JS}(P_{\text{base}}, P_{\text{condition}})$ is averaged with a task-block bootstrap.

The contrast is topological. In ReAct, the scratchpad matters only when the task needs it. In the multi-agent deployment, the controller’s decision is routed through the private worker report, so the active site is the communication edge itself. Same-class variation is local, neutral removal is substantial, and counterfactual evidence reaches 0.901 bits, close to the oracle ceiling.

7 Discussion

7.1 One Plot: Capacity, Then Activation

The three experiments follow a single logic. Structural budgets map where hidden capacity can reside—scratchpad modules, denoising interfaces, peer-state edges. Dynamic certificates identify which of those sites affect actions. The same scratchpad is dormant on calculator tasks and active on planning tasks in ReAct; in LLaDA, action relevance concentrates at the final binding step; in the multi-agent deployment, delegation makes the private worker report the active edge.

Effect sizes index deployment topology. Under the same ReAct topology, calculator tasks leave the scratchpad dormant while planning tasks produce nonzero action-distribution shifts under intervention (0.0215 bits, CI [0.0094, 0.0410]) and controlled replay (0.0163 bits, CI [0.0124, 0.0208]). The LLaDA final binding step rises to 0.110 bits; the multi-agent counterfactual report edge rises to 0.901 bits. A hidden channel that supplements an already-determined visible trace has low $\delta_{\text{act}}^{\text{LB}}$; one that

Condition	$\delta_{\text{act}}^{\text{LB}}$ (JS bits)	95% CI (bits)
Neutral replay	0.358	[0.324, 0.392]
Counterfactual evidence	0.901	[0.873, 0.928]
Same-class shuffle	0.084	[0.060, 0.109]
Oracle tag upper-bound	1.000	[1.000, 1.000]

Table 4: Topological activation profile for the multi-agent deployment. Counterfactual evidence nearly saturates the binary action ceiling, while same-class shuffling remains much smaller. Hidden communication is therefore behaviorally active when the controller has delegated evidence gathering to the worker.

carries the decision has high $\delta_{\text{act}}^{\text{LB}}$. Under this protocol, a claim that an unobserved variable causally affects behavior must identify both a channel in $\varepsilon_{\text{state}}^{\text{UB}}$ and a probe in $\delta_{\text{act}}^{\text{LB}}$.

Acknowledgments

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366 **NeurIPS Paper Checklist**

367 **1. Claims** Do the main claims made in the abstract and introduction accurately reflect the paper’s
368 contributions and scope?

369 *Answer:* Yes. The abstract and introduction state the dual-certificate framework, the structural upper
370 budget (Proposition 1), the dynamic lower bound (Proposition 2), and the experimental results that
371 correspond to them. All claims are matched by theoretical derivations (§4–5) and empirical evidence
372 (§6).

373 **2. Limitations** Does the paper discuss limitations of the work?

374 *Answer:* Yes. Limitations are discussed in several places: the observational bottleneck and proxy-
375 estimator bias/variance (Appendix C), the scope of claims (§7), the conservative nature of the
376 structural budget, and the dependence on software-orthogonality assumptions for tightness (§4,
377 Appendix A).

378 **3. Theory Assumptions and Proofs** Did you state the full set of assumptions of all theoretical
379 results, and did you include complete proofs of all theoretical results?

380 *Answer:* Yes. All assumptions are stated with each proposition (software orthogonality in Lemma 1,
381 conditional Markov and DPI for Proposition 2). Complete proofs are given in the main text (proof
382 sketches) and in full detail in Appendix A.

383 **4. Experimental Result Reproducibility** Did you provide a way to reproduce the experimental
384 results?

385 *Answer:* Yes. Detailed experimental protocols are described in Appendix B, including the structural-
386 budget computation procedure and the dynamic-certificate estimation steps (proxy, replay, interven-
387 tion, and private-report intervention). The paper references implementation README files. Code
388 and data are included in the supplementary material.

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390 reproduce the main experimental results?

391 *Answer:* Yes. Code, precomputed results, and reproduction instructions are provided in the supple-
392 mentary material. The experiments use the open-weight Qwen2.5-7B-Instruct model (Apache 2.0).

393 **6. Experimental Setting/Details** Did you specify all the training details (e.g., data splits, hyperpa-
394 rameters, how they were chosen)?

395 *Answer:* Yes. Experimental details are reported in §6 and Appendix B: e.g., 5-fold cross-fitted logistic
396 regression, trajectory-block bootstrap with 95% CIs, PCA dimensions, audit discretization $Q = 256$,
397 perturbation levels, controller sample count k , and task-pool sampling seed. Hyperparameter choices
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419 limitations and the conservative nature of the bounds (§7). The overall contribution promotes
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427 supporting software communities (Qwen2.5, LLaDA, Hugging Face Transformers, PyTorch, NumPy,
428 scikit-learn, NetworkX, and Matplotlib; §* Acknowledgments), and uses publicly available assets in
429 accordance with their respective licenses.

430 **13. New Assets** Are you releasing new assets (datasets, code, models), and did you document them
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432 *Answer:* No. The experimental code and data are provided in the supplementary material for review
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442 *Answer:* N/A. The LLM (Qwen2.5) serves only as the audit target—it is not a component of the
443 proposed auditing method. The auditing framework itself (min-cut computation, mutual-information
444 lower bounds, and the dual-certificate logic) is independent of any particular language model and
445 does not rely on LLMs as a core methodological component.

A Network Information Theory Background for Proposition 1

Conditional channel capacity. For each unlogged edge e in the time-unrolled DAG G_t , the conditional capacity c_e is the maximum information that can flow through e under the worst-case distribution consistent with the system protocol. It is not an empirical average mutual information. It is a structural upper bound read from interface specifications: token window length, hidden-dimension size, quantization level, and discrete-state cardinality. Concretely, a text channel with budget K tokens over vocabulary $\mathcal{V}$ has $c_e \leq K \log |\mathcal{V}|$; a vector state of dimension d at quantization Q has $c_e \leq d \log Q$; a discrete scheduler with $|\Omega|$ states has $c_e \leq \log |\Omega|$. The $K \log |\mathcal{V}|$ form is the maximum-entropy bound (uniform over K -token sequences); any other distribution gives strictly lower entropy. Auditors who can measure the empirical input distribution to a particular channel may substitute a tighter bound, which only improves the certificate.

What tightening means. The structural budget remains valid under any replacement $c_e^* \geq I(X_e; Y_e \mid \tilde{T}_t)$, so tightening is a matter of choosing smaller but still defensible per-edge budgets. In practice, this means replacing the worst-case $K \log |\mathcal{V}|$ or $d \log Q$ budget with an empirical or conditional estimate at the same audit discretization, rather than changing the theorem itself.

Cut-set upper bound and additive decomposition. Let $G_t = (V_t, E_t)$ be the time-unrolled DAG. For a cut $\Omega \subseteq V_t$ separating source nodes from a target node, the *induced cut capacity* $C_{\text{cut}}(\Omega)$ is the supremum, over joint input distributions $p(X_\Omega \mid \tilde{T}_t)$, of the directed conditional mutual information $I(X_\Omega \rightarrow Y_{\Omega^c} \mid \tilde{T}_t, X_{\Omega^c})$. The network-information-theoretic cut-set upper bound [El Gamal and Kim, 2011] states that any information flowing from sources in Ω to a target in Ω^c is bounded above by $C_{\text{cut}}(\Omega)$, and minimizing over cuts gives the tightest such bound. Proposition 1 applies this bound to the unrecorded trace fragment M_t . We use the cut-set *upper bound* direction only; the network-coding achievability results [Ahlsweide et al., 2000], which concern lower bounds on capacity under cooperative coding across the cut, are not invoked. The induced capacity $C_{\text{cut}}(\Omega)$ is in general a supremum over joint input distributions and cannot be read directly from interface specifications. Lemma 1 shows that under software orthogonality, when the channel laws across the cut factor into independent single-edge channels conditional on $\tilde{T}_t$, the induced capacity is bounded above by the sum of per-edge capacities:

$$C_{\text{cut}}(\Omega) \leq \sum_{e \in E(\Omega, \Omega^c)} c_e.$$

Corollary 1 combines this with Proposition 1 to reduce the certificate computation to a discrete minimum-cut problem on G_t with edge capacities c_e . This is the form used in the structural-budget experiment and in the protocol in §B.

Formal derivation of Proposition 1. The proof in §4 has two steps: a trace-gap identity and a cut-set bound via d -separation. We restate the argument here with the d -separation step explicit.

Step 1: Trace-gap identity. Let $M_t := T_t \setminus \tilde{T}_t$. The chain rule of conditional entropy gives

$$H(S_t \mid \tilde{T}_t) = H(S_t \mid T_t) + I(S_t; M_t \mid \tilde{T}_t).$$

The first term is $\varepsilon_{\text{nominal}} = H(S_t \mid T_t)$.

Step 2: Cut-set bound via d -separation. Fix a cut $\Omega \in \text{Cuts}(U_t \rightarrow S_t)$. By the definition of a cut in the time-unrolled DAG G_t , every directed path from a node in Ω to $S_t \in \Omega^c$ must cross at least one edge in $E(\Omega, \Omega^c)$. The causal influence of M_t on S_t must be mediated by the cross-cut message stream $(X_\Omega \rightarrow Y_{\Omega^c})$ together with the boundary state X_{Ω^c} . Formally, M_t is d -separated from S_t by $(X_\Omega \rightarrow Y_{\Omega^c}, X_{\Omega^c})$ conditional on $\tilde{T}_t$, which yields

$$I(S_t; M_t \mid \tilde{T}_t) \leq I(X_\Omega \rightarrow Y_{\Omega^c} \mid \tilde{T}_t, X_{\Omega^c}).$$

By the definition of $C_{\text{cut}}(\Omega)$ as the supremum of the right-hand side over admissible joint input distributions on X_Ω ,

$$I(S_t; M_t \mid \tilde{T}_t) \leq C_{\text{cut}}(\Omega).$$

488 Since this holds for every cut, it holds for the minimum,

$$I(S_t; M_t \mid \tilde{T}_t) \leq \min_{\Omega \in \text{Cuts}(U_t \rightarrow S_t)} C_{\text{cut}}(\Omega).$$

489 Combining with Step 1 completes the proof of Proposition 1. Corollary 1 follows immediately by sub-
 490 stituting Lemma 1 into each software-orthogonal cut: under orthogonality, $C_{\text{cut}}(\Omega) \leq \sum_{e \in E(\Omega, \Omega^c)} c_e$,
 491 and the minimum over cuts becomes a discrete min-cut problem on the unlogged subgraph with edge
 492 capacities c_e .

493 B Experimental Protocols

494 B.1 Structural Budget Computation Protocol

495 Inputs: deployment architecture specification, logging manifest, and protocol budgets (token window
 496 sizes, hidden dimensions, quantization levels). Procedure: (1) construct the time-unrolled DAG G_t ;
 497 (2) partition edges into logged and unlogged; (3) verify software orthogonality (Lemma 1) or add
 498 shared-state edges explicitly to G_t until orthogonality holds; (4) assign capacity c_e per unlogged
 499 edge using the forms in §4; (5) enumerate reachable cut sets $\mathcal{C}_{\text{unlogged}}$ with a min-cut algorithm after
 500 logged edges are collapsed to zero residual capacity; (6) compute $\varepsilon_{\text{state}}^{\text{UB}} = \varepsilon_{\text{nominal}} + \min_C \sum_{e \in C} c_e$
 501 via Corollary 1.

502 For the logging ablation, repeat steps (2)–(6) for the ordered manifests used in §6.1: output-
 503 only trace, tool-call trace, router-logit trace, summary-buffer trace, and full scratchpad
 504 trace. Report the upper-bound value and active minimum-cut edges at each level. See
 505 experiments/7.1_static_certificate/README.md for implementation.

506 B.2 Dynamic Certificate Estimation Protocol

507 **Proxy certificate.** Inputs: model, task environment, and probe family $Z_t^{(k)} = f_k(S_t)$. Steps:
 508 (1) run inference on N trajectories and record $(\tilde{T}_t, Z_t^{(k)}, A_t)$; (2) train cross-fitted predictors $p_\theta(A_t \mid$
 509 $\tilde{T}_t)$ and $p_\phi(A_t \mid \tilde{T}_t, Z_t^{(k)})$; (3) compute the cross-entropy difference $\hat{I}_k = L_{\text{trace}} - L_{\text{trace}+Z^{(k)}}$;
 510 (4) estimate trajectory-block bootstrap intervals; (5) repeat for random, label-permuted, 1-dimensional,
 511 3-dimensional, 5-dimensional, and full-slice proxies.

512 **Optional controlled replay certificate.** Inputs: model and two controlled execution regimes (wild
 513 and replay). Steps: (1) for each $\tilde{T}_t$, run the same system under wild and replayed missing-state
 514 conditions; (2) estimate $\hat{\Delta} = \hat{\text{JS}}(P_{\text{wild}}, P_{\text{replay}} \mid \tilde{T}_t)$; (3) aggregate over trace slices. This is not
 515 output-only paraphrase replay; it is valid only under the controlled replay condition in §5.

516 **Intervention certificate.** Inputs: model, targeted hidden module, perturbation budget, and task split.
 517 Steps: (1) inject ξ_{hidden} at levels $\{0, \sigma_1, \sigma_2, \dots\}$ while holding $\tilde{T}_t$ fixed; (2) record $(\xi_{\text{hidden}}, A_t \mid \tilde{T}_t)$
 518 pairs; (3) estimate $\hat{I}(\xi_{\text{hidden}}; A_t \mid \tilde{T}_t)$ via conditional action-distribution shift for discrete action
 519 spaces, or via InfoNCE/MINE for continuous action spaces; (4) repeat on dormant tasks and active
 520 tasks under the same topology to test whether $\delta_{\text{act}}^{\text{LB}}$ changes while $\varepsilon_{\text{state}}^{\text{UB}}$ remains fixed.

521 **Multi-agent private-report certificate.** Inputs: task pool, worker model, controller model, and a
 522 two-class API space. Steps: (1) sample $N/2$ arithmetic tasks and $N/2$ factual-retrieval tasks; (2) have
 523 the worker generate a private evidence report under a prompt that forbids direct API names and audit
 524 each report for leakage; (3) sample the controller action distribution k times under wild, neutral
 525 replay, opposite-class report, same-class shuffled report, and oracle-tag conditions; (4) parse actions
 526 into $\{\text{calculator}, \text{search}, \text{invalid}\}$; (5) compute per-task JS divergence in bits and task-block
 527 bootstrap intervals. The oracle-tag contrast is reported only as an upper bound; the main contrast is
 528 the opposite-class worker-report intervention.

529 See the README files for experiments 7.2, 7.3, and 7.6 for implementation.

C Proxy Estimator: Bias and Variance

The read-only proxy experiment is a calibration of observational access, not a main activation result. On 3,000 tool-selection trajectories (mixed calculator/search/email/calendar/weather), we record $\tilde{T}_t$ (embedding-layer representation), Z_t (layer-24 tool-logit projection), and A_t . The CE-diff $\hat{I} = \mathcal{L}(A_t | \tilde{T}_t) - \mathcal{L}(A_t | \tilde{T}_t, Z_t)$ is estimated via 5-fold cross-fitted logistic regression with trajectory-block bootstrap. Table 5 reports the resolution ablation: random, 1/3/5-dim PCA, and full 10-dim proxy.

The $d=5$ PCA proxy recovers 0.0317 bits (1.4% of the $\log_2 5 = 2.32$ bit ceiling) with CI excluding zero on the mixed-task distribution. The magnitude reflects an *observational bottleneck*: when the baseline predictor achieves near-perfect accuracy, the ceiling $H(A_t | \tilde{T}_t) \approx 0$ caps any read-only proxy. Intervention and replay avoid this bottleneck by changing the missing-state condition while holding the visible transcript fixed.

The dormant/active proxy split should be read as an estimator-noise diagnostic. On calculator tasks (dormant scratchpad), $H(A_t | \tilde{T}_t) = 0$ empirically, yielding $\delta_{\text{act}}^{\text{LB}} = 0$ trivially. On planning tasks (active scratchpad), the proxy row is positive in point estimate but has a wide CI that straddles zero. We therefore do not use the read-only proxy as the main body evidence for active scratchpad use; the causal intervention and controlled replay in §6.2 supply that evidence.

Proxy	CE-diff (bits; nats)	95% CI (bits)
<i>Resolution ablation (mixed task)</i>		
random	−0.0007; −0.0005	[−0.0012, −0.0003]
$d = 1$ PCA	+0.0065; +0.0045	[+0.0053, +0.0078]
$d = 3$ PCA	+0.0250; +0.0173	[+0.0225, +0.0273]
$d = 5$ PCA	+0.0317; +0.0220	[+0.0273, +0.0358]
full (10-dim)	+0.0183; +0.0127	[+0.0134, +0.0229]
<i>Dormant/active split ($d=64$ PCA, L2-regularized)</i>		
calculator (dormant)	0.0000; 0.0000	[0, 0]
planning (active)	+0.0412; +0.0286	[−0.0725, +0.1386]

Table 5: Read-only proxy calibration. Top: resolution ablation on the mixed tool-selection dataset. Bottom: dormant/active split. The active-task row is reported honestly as an estimator-noise diagnostic because its confidence interval straddles zero; causal replay/intervention supplies the body evidence for active scratchpad use. Negative random-control CE-diff is a finite-sample artifact near the noise floor and is clipped to zero when interpreted as a lower-bound certificate.

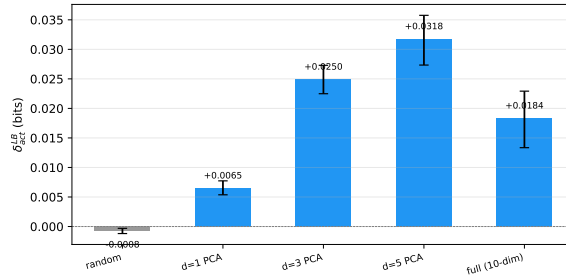


Figure 3: Proxy resolution ablation. $\delta_{\text{act}}^{\text{LB}}$ increases monotonically through $d=5$ PCA. The random proxy sits near zero (estimator leakage check). Error bars are 95% trajectory-block bootstrap CIs.

The CE-diff estimator $\hat{I} = L(\tilde{T}_t) - L(\tilde{T}_t, Z_t)$ lower-bounds $I(Z_t; A_t | \tilde{T}_t)$, but three bias sources and two variance sources matter in practice.

Bias sources. (1) *Proxy coarsening*: $Z_t = f(S_t)$ discards information in S_t not captured by f , so $I(Z_t; A_t | \tilde{T}_t) \leq I(S_t; A_t | \tilde{T}_t)$. The gap is the coarsening loss. (2) *Restricted predictor class*: Logistic regression or shallow MLPs may not reach the Bayes-optimal predictor, so the CE-diff

underestimates the true conditional MI. (3) *Finite-sample bias*: With finite N , the cross-entropy on held-out folds is an upward-biased estimate of the population CE, and the difference of two such estimates carries residual bias. Cross-fitting mitigates but does not eliminate this.

Variance sources. (1) *Trajectory sampling*: The N observed trajectories are a finite sample from the deployment distribution. Block bootstrap over trajectories estimates the resulting CI. (2) *Proxy dimension*: Higher-dimensional proxies carry more information but increase predictor variance. The PCA ablation in Table 5 shows this trade-off: $d=5$ outperforms both $d=3$ and the full 10-dimensional proxy.

Mean squared error. Decomposing $\text{MSE} = (\text{bias})^2 + \text{variance}$, the proxy certificate is a conservative lower bound: positive bias (underestimation) is safe for the audit (it never overstates δ_{act}), while variance widens the CI but does not shift the point estimate.

Sampling noise and temperature. Increasing decoding temperature can raise $H(A_t | \tilde{T}_t)$, but independent sampler noise is not a certificate amplifier. It changes the audited policy and can reduce $I(Z_t; A_t | \tilde{T}_t)$ when the added action variation is unrelated to the hidden state proxy. A temperature sweep is therefore a calibration of a randomized deployment policy, not a substitute for proxy quality or causal access. We do not use temperature injection for the main proxy certificate; the synthetic experiment instead injects hidden-state stochasticity with known ground truth.

D Synthetic Ground-Truth Validation

To test whether the certificates recover known hidden influence, we construct a synthetic agent whose hidden-state entropy and decision relevance are controlled by design. A binary hidden variable $H \sim \text{Bern}(0.5)$ injects influence of strength β_h into the action logits; the unlogged-channel capacity $c_H = H(H) = 1$ bit is known. We run both the static min-cut ($\varepsilon_{\text{state}}^{\text{UB}} = 1$ bit) and the proxy CE-diff estimator on 1,000 trajectories at five influence levels.

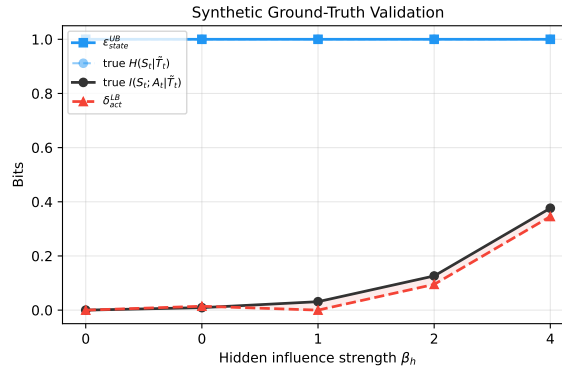


Figure 4: Synthetic ground-truth. $\varepsilon_{\text{state}}^{\text{UB}}$ (blue) equals the true $H(S_t | \tilde{T}_t) = 1$ bit. $\delta_{\text{act}}^{\text{LB}}$ (red) tracks the true $I(S_t; A_t | \tilde{T}_t)$ from below, with negative values at low β_h reflecting estimator variance near the noise floor.

Fig. 4 shows $\varepsilon_{\text{state}}^{\text{UB}} = 1$ bit matches the true hidden entropy (the bound is tight when there is no redundant unlogged capacity). The dynamic lower bound $\delta_{\text{act}}^{\text{LB}}$ tracks the true conditional MI from below at all tested influence levels ($\beta_h \geq 1$); negative CE-diff values at $\beta_h = 0$ reflect finite-sample variance around a near-zero true MI.

The Qwen deployment has a near-deterministic baseline that suppresses the proxy signal ($H(A_t | \tilde{T}_t) \approx 0$). The synthetic environment instead injects controlled stochasticity ($H = 1$ bit of hidden-state entropy). Without that observational bottleneck, the proxy certificate $\delta_{\text{act}}^{\text{LB}}$ tracks the true conditional-MI curve from below in the tested range. The validation shows that the proxy certificate can become tight when structural capacity is behaviorally active *and* the observational ceiling permits detection.

E Tightening the Static Certificate

The reported structural budget is intentionally conservative. Its role is to upper-bound the residual hidden-state entropy from deployment topology alone, so the default budgets in §4 are worst-case interface capacities rather than empirical information estimates.

Empirical and conditional budgets. A tighter deployment-specific certificate can replace the worst-case budget c_e on each unlogged edge by an empirically justified budget c_e^* . For a text channel, one option is a high-confidence upper bound on $H(X_e \mid \tilde{T}_t)$; when the visible history strongly constrains the missing input, a conditional form such as $H(X_e \mid \tilde{T}_t, \mathcal{H}_t)$ can be smaller still. For quantized vector states, the analogous replacement is an empirical entropy or rate-distortion budget at the same audit discretization. These substitutions do not alter the min-cut proof: they only change the edge weights used by Corollary 1.

Graph refinement. If two apparent channels share latent state, naive edge-wise additivity can double-count the same information. The remedy is to refine the time-unrolled DAG so that the shared state appears as an explicit node or edge family, then recompute the min-cut on the refined graph. This usually tightens the certificate more reliably than trying to compensate with ad hoc weight corrections.

Reporting policy. For review and deployment, report a ladder of certificates: worst-case structural, empirical-marginal, and empirical-conditional. The first is the theorem-backed safety baseline. The latter two are optional tightening modes and should be reported only when the underlying measurements are defensible.